

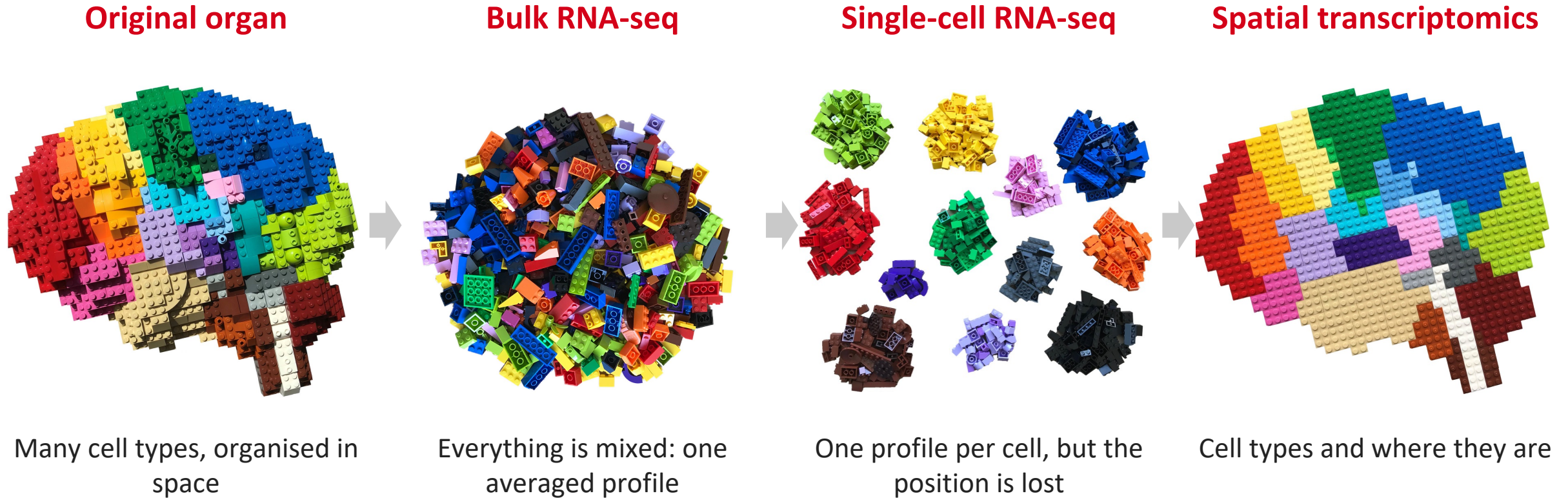
Spatial (and single-cell) transcriptomics of human skin wound healing

ENS de Lyon · NGS practicals 2026

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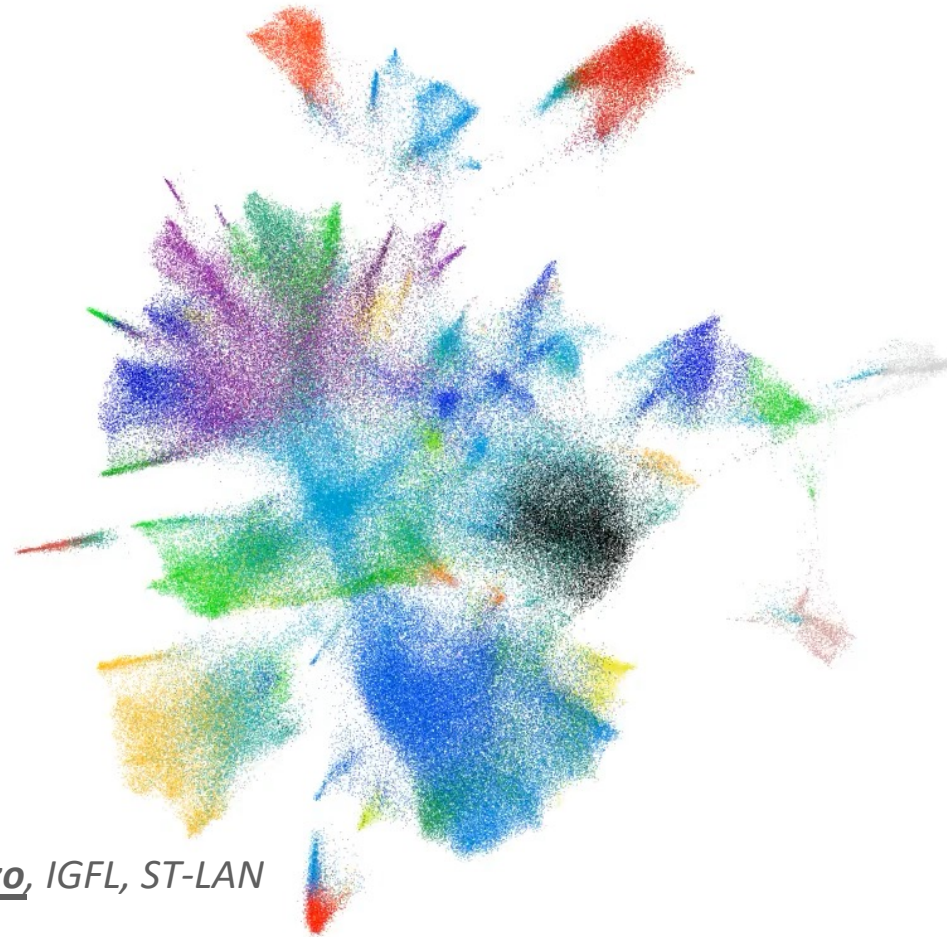


Adding new dimensions to transcriptomics



Today: we use both on the same wound, to ask which cells are present (single-cell) and where they sit (spatial).

Adding new dimensions to transcriptomics



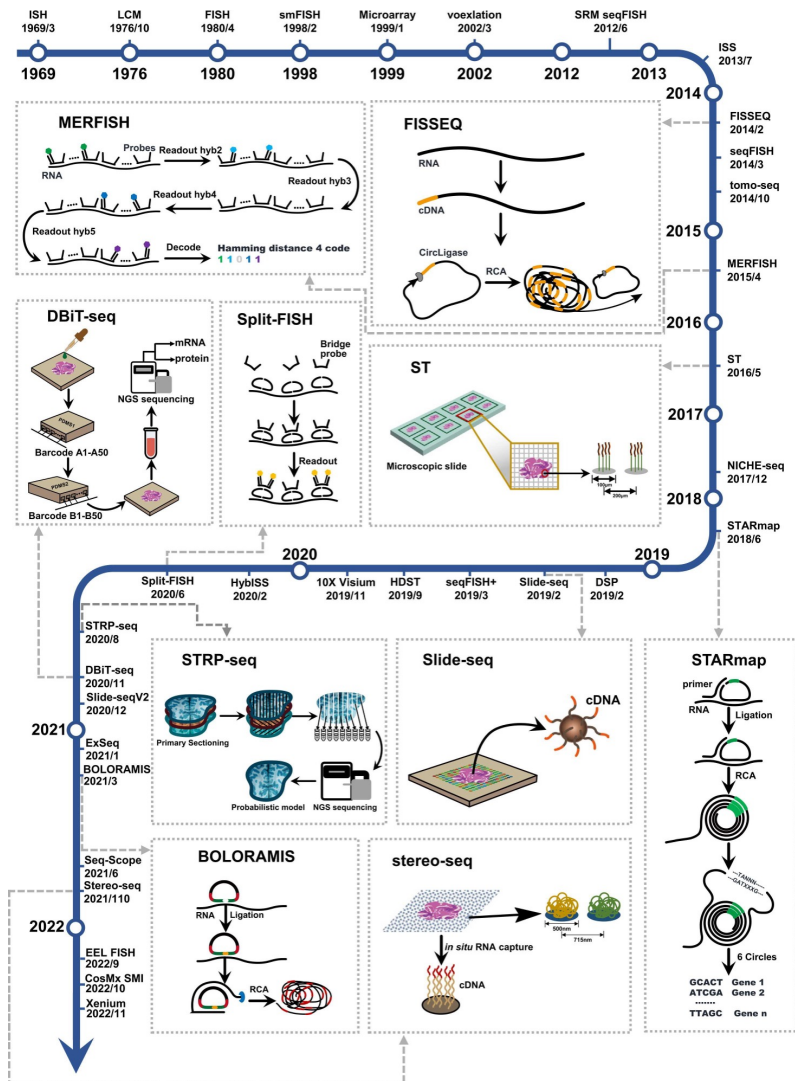
Unpublished (true) data from Marco Uderzo, IGFL, ST-LAN
group: VISIUM HD from Mouse Brain

Spatial transcriptomics in 2026: a crowded field

Technology	Approach	Resolution	Genes
Visium	Sequencing, barcoded array	55 µm spots (several cells)	Whole transcriptome
Visium HD	Sequencing, barcoded array	2 µm features	Whole transcriptome
Stereo-seq	Sequencing, DNA nanoballs	about 0.2 µm features	Whole transcriptome
MERFISH	Imaging	Single cell / subcellular	Targeted panel
Xenium	Imaging	Single cell / subcellular	Panel, up to about 5,000 genes
CosMx SMI	Imaging, optical barcodes	Single cell / subcellular	Panel, up to about 6,000 genes

Our data: classic Visium. Each 55 µm spot mixes several cells, so we will need single-cell data to say which cells are in each spot.

The long journey of spatial transcriptomics



Microdissection

Tissues are dissociated and the different parts are sequenced

In situ hybridisation (ISH)

Fluorescence microscopy tags transcripts inside cells

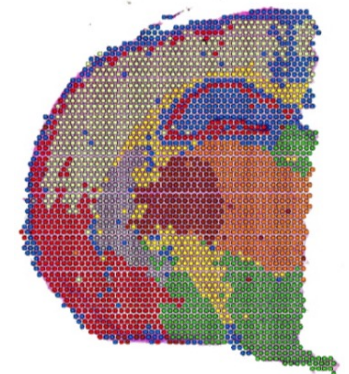
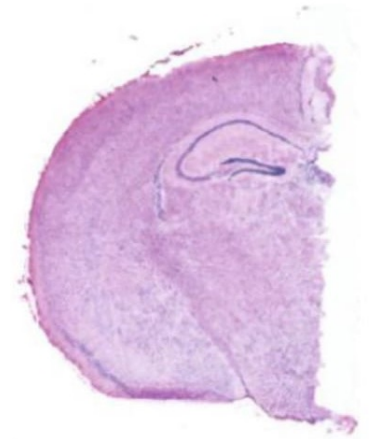
In situ sequencing (ISS)

Short cDNA fragments are sequenced in situ

In situ capture (ISC)

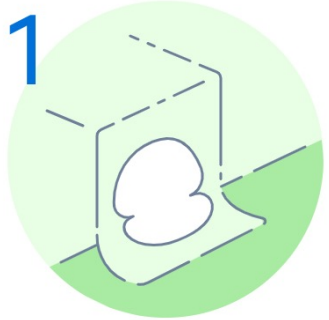
Transcripts from a tissue section are captured on a barcoded array: Visium

Tissue section



Spatially resolved expression

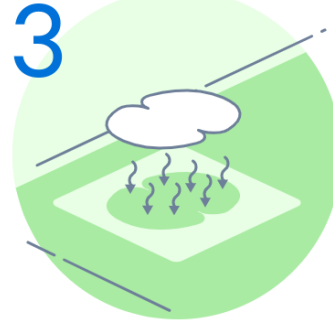
Visium workflow, step by step



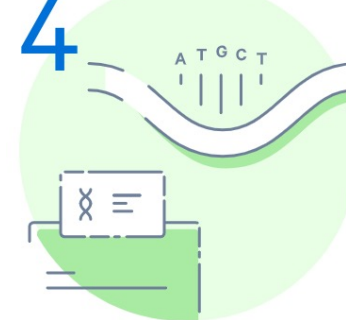
Sample preparation



Tissue imaging (H&E)



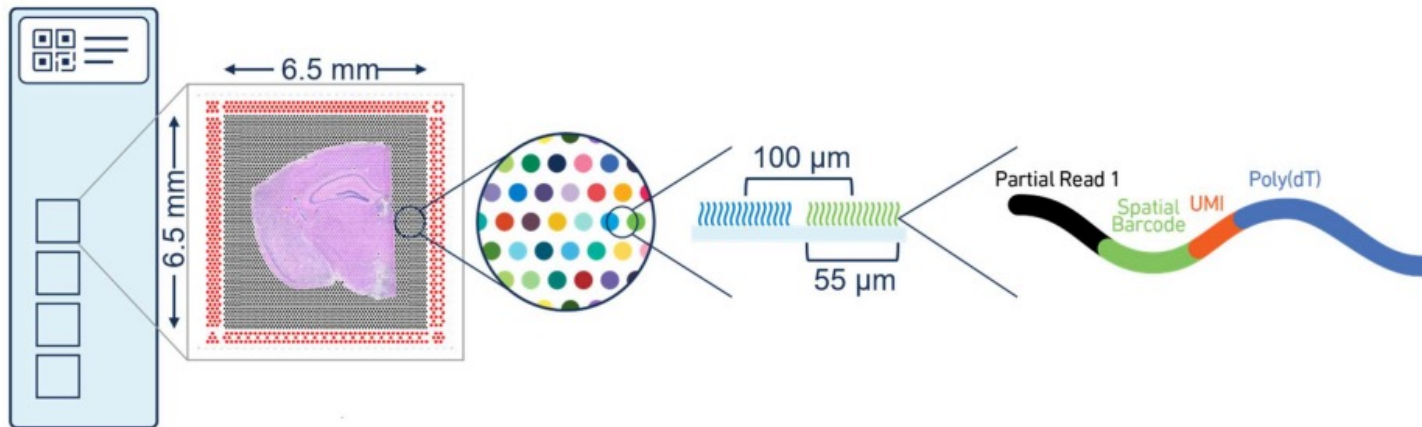
Library construction



Sequencing



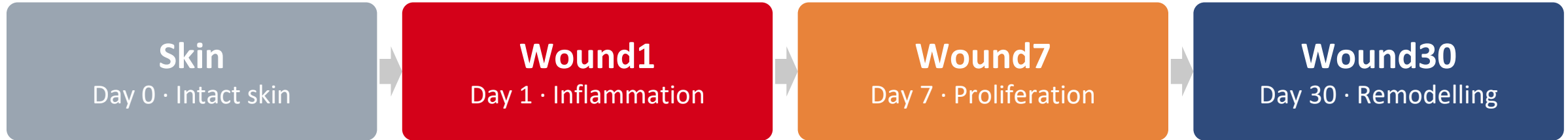
Data analysis



What a spot gives

- Spatial barcode: position on the slide
- UMI: one count per molecule
- 55 µm spot: several cells

The case study: human skin wound healing



- Healthy volunteers, 4-mm full-thickness wounds on the upper buttock
- Wound-edge biopsies at day 1, 7 and 30, plus intact skin
- Each donor is their own control over time
- Question: which cells enter the wound, when, and where?

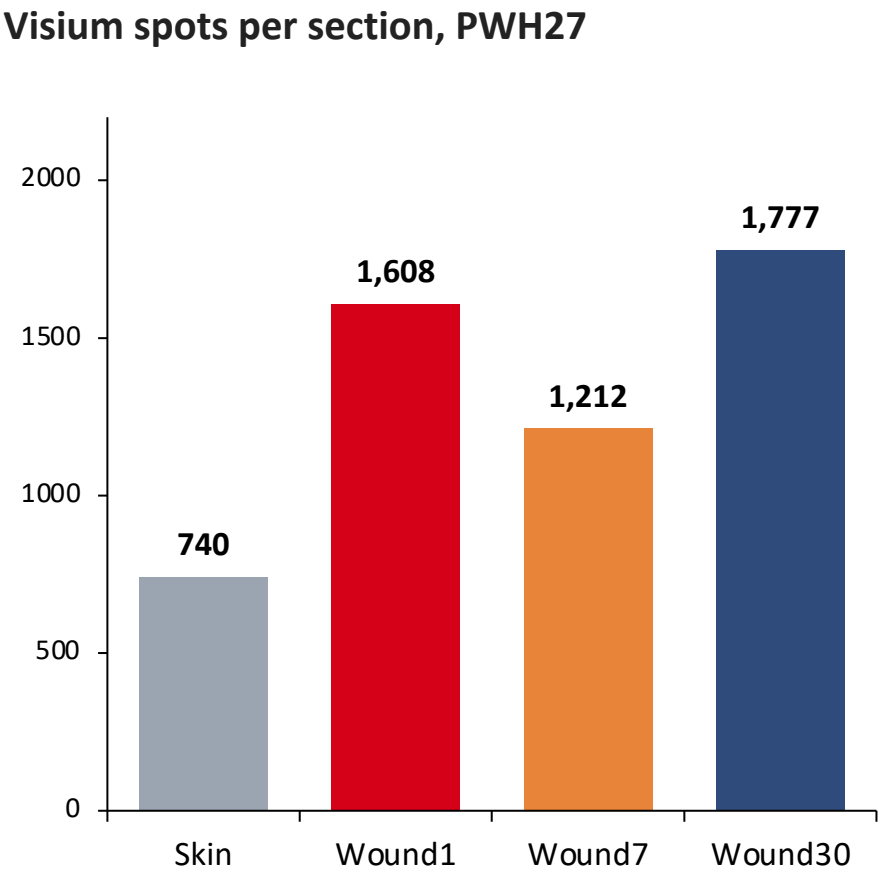
Why it matters

Failed re-epithelialisation is a hallmark of chronic wounds (venous and diabetic foot ulcers), which affect millions of people.

Two datasets, same wounds

	Single-cell (Chromium)	Spatial (Visium)
GEO	GSE241132	GSE241124
Donors	3 (PWH26, 27, 28)	4 (PWH19, 20, 27, 28)
Samples	12 (3 donors × 4 time points)	16 sections (4 × 4)
Size	58,823 cells	22,915 spots
Our subset	PWH27: 22,227 cells	PWH27: 5,337 spots

PWH27 is in both datasets: the same individual, the same four time points, in single-cell and in space.



What the paper found

1 Migrating keratinocytes

Basal and spinous migrating clusters peak at Wound1; FOSL1 emerges as a driver of migration.

2 A relay between cell types

Pro-inflammatory macrophages support migration early; proliferating fibroblasts take over at day 7.

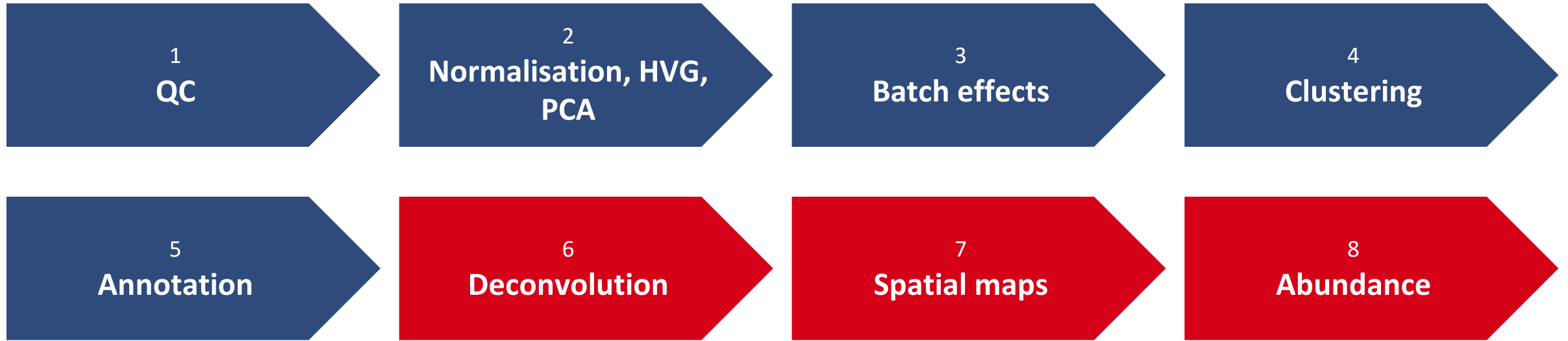
3 Fibroblast states shift

Mesenchymal fibroblasts (POSTN+) increase during healing and remain elevated up to day 30.

4 Space matters

Cell2location maps the single-cell types onto Visium spots: a wound-margin structure is visible in the tissue.

The practicals: from raw data to spatial maps



Part 1 · Visium and single-cell, each on its own

- Same steps on spots and on cells
- scanpy, squidpy, spatialdata
- Cluster resolution chosen with RESOLUTE

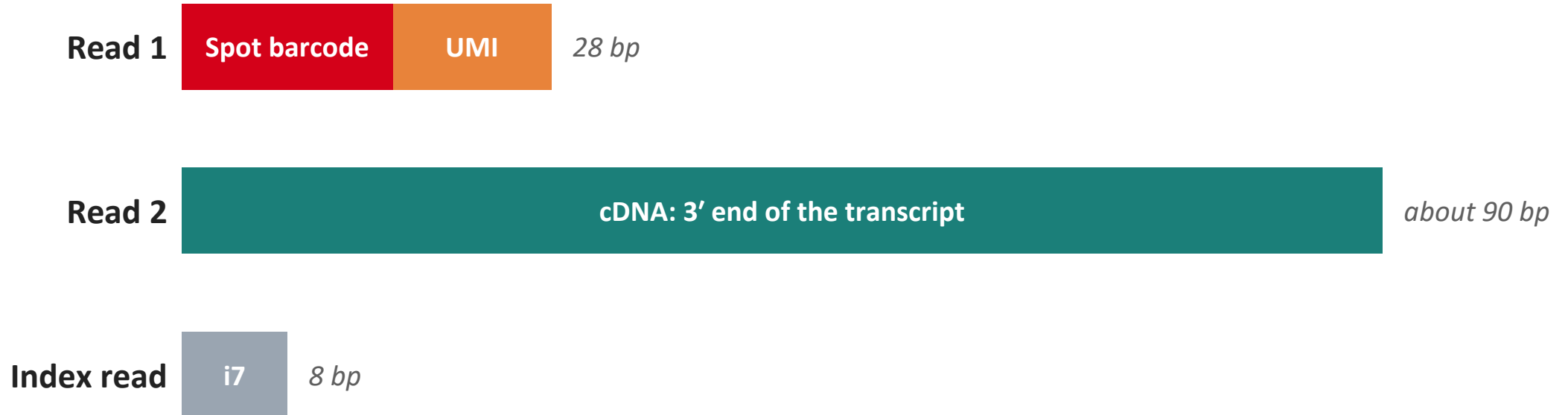
Part 2 · Bringing the two together

- Cell2location: single-cell signatures → cell types per spot
- Compare time points, as the paper does

Setting up

Alignment: from reads to counts

What is in the FASTQ files: Visium reads



Spot barcode

Which spot (position on the slide) did this molecule come from?

UMI

Which original mRNA molecule?
Let's collapse PCR duplicates.

cDNA

Which gene? Mapped to the reference genome.

Sample index

Which library? Demultiplexes pooled samples.

Alignment: where does each read come from?



Why align?

- Each R2 read is placed on the reference genome (STAR)
- Reads on an exon are assigned to a gene; barcode + UMI give spot and molecule

Space Ranger (10x Genomics)

- One command, spaceranger count, does all the steps above
- The reference is built once beforehand with spaceranger mkref

Output: a spots × genes count matrix and the aligned tissue image: the starting point of our notebook.

We do not have time to do a real alignment. We will do a mock alignment and use aligned reads from the paper.

What spaceranger count needs

1 FASTQ files

Read 1 (spot barcode + UMI) and read 2 (cDNA) for each sample, as delivered by the sequencer.

2 A reference

Genome FASTA and gene annotation (GTF), indexed once. For the human genome this takes hours and a lot of RAM.

3 The tissue image

Microscope image of the capture area: used to find the fiducial frame and the spots under tissue.

4 The output

Count matrix (filtered_feature_bc_matrix.h5), spot positions and the aligned image in spatial/.

Today: a mock alignment

What we will do

- Run spaceranger count on tiny toy FASTQ files for our 4 samples (Skin, Wound1, Wound7, Wound30)
- Read the command line, the logs and the web_summary.html
- Reference already built: we only run count

What we will not do

- Align the real data: too long for a practical
- Interpret the toy results: the sequences are random
- Build the reference ourselves

Our plan: run the mock alignment, then ignore its output and start the notebook from the real, already aligned data (GSE241124).

À vous de jouer !

Open the first notebook and have fun!

References

- Liu Z. et al. Spatiotemporal single-cell roadmap of human skin wound healing. *Cell Stem Cell* 32, 479–498 (2025).
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- Lim H.J., Wang Y., Buzdin A., Li X. A practical guide for choosing an optimal spatial transcriptomics technology. *BMC Genomics* 26, 47 (2025).
- Kleshchevnikov V. et al. Cell2location maps fine-grained cell types in spatial transcriptomics. *Nat Biotechnol* 40, 661–671 (2022).
- Wolf F.A., Angerer P., Theis F.J. SCANPY. *Genome Biol* 19, 15 (2018); Palla G. et al. Squidpy. *Nat Methods* 19, 171–178 (2022).